NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 61**

REVISION ASSINGMENT # 15 (APPLICATION OF DERIVATIVES-01) TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. Let $f(x) = x^3 + ax + b$ with $a \neq b$ and suppose the tangent lines to the graph of f at $x = a$ and $x = b$ have the same gradient. Then the value of $f(1)$ is equal to -

(A) 0 (B) 1 (C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

2. If the points of local extremum of $f(x) = x^3 - 3ax^2 + 3(a^2 - 1)x + 1$ lies between –2 & 4, then 'a' belongs to -

(A) (–2,2) (B) $(-\infty, -1) \cup (3, \infty)$ (C) (–1, 3) (D) $(3, \infty)$

3. Let f be a function which is continuous and differentiable for all real x. If $f(2) = -4$ and $f'(x) \geq 6$ for all $x \in [2, 4]$ then

(A) $f(4) < 8$ (B) $f(4) \geq 8$ (C) $f(4) \geq 12$ (D) none of these

4. If x and y are real numbers such that $x^2 + y^2 = 27$, then maximum possible value of x – y is -

(A) $\sqrt{72}$ (B) $\sqrt{18}$ (C) $\sqrt{54}$ (D) $\sqrt{\frac{2}{3}}$

5. A boat leaves the dock at 2 PM and travels due south at a speed of 20 km/h. Another boat has been heading due east at 15 km/hr and reaches the same dock at 3 PM. At what time the two boats were closest to each other -

(A) 2:21:36 PM (B) 2:15:00 PM (C) 2:15:15 PM (D) 2:21:40 PM

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6. Consider the function for $x = [-2, 3]$, $f(x) = \begin{cases} \frac{x^3 - 2x^2 - 5x + 6}{x - 1} & \text{if } x \neq 1 \\ -6 & \text{if } x = 1 \end{cases}$, then

(A) f is discontinuous at $x = 1 \Rightarrow$ Rolle's theorem is not applicable in [–2, 3]
(B) $f(-2) \neq f(3) \Rightarrow$ Rolle's theorem is not applicable in [–2, 3]
(C) f is not derivable in (–2, 3) $\Rightarrow$ Rolle's theorem is not applicable
(D) Rolle's theorem is applicable as f satisfies all the conditions and c of Rolle's theorem is 1/2

7. If $x.y = 1$, then minimum value of $x^2 + y^2$ is -

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) 4

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

8. If $f(x) = 3x^2 + \int_0^x e^{-t} f(x - t)dt$, then

(A) $f(x) = 0$ has 3 solutions (B) $f(x)$ is monotonic increasing

(C) $f(x) = 4$ has 2 solutions (D) $\int_{-1}^{1} f(x)dx = 2$

9. Let f be a differentiable function $\forall\ x \in R$ where $f(1) = 1$ & $f(3) = 4$ & $f'(x) \geq 1\ \forall\ x \in R$, then $f(2)$ can be equal to

(A) 1 (B) 2 (C) 3 (D) 4

10. Let $f(x)$ is a derivable function which is increasing for all $x \in R$ (having no critical point), then

(A) $f(3 - 4x)$ is an increasing function for all x. (B) $f(3 - 4x)$ is a decreasing function for all x.

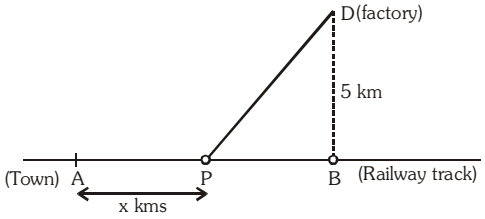
(C) $f(x^2 - x)$ increases for $x > \frac{1}{2}$. (D) $(f(x))^3$ is an increasing function for all x.

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Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 11 to 12

A factory is connected with a straight railway track by a highway. Town A lies on the railway track. The distance DB from the factory to railway track is 5 km. The segment AB of railway track is equal to 'ℓ' kms. Freight charges on highway are 2 times higher than on railway. A highway DP is connected to railway track such that AP is x kms (see the figure).



11. If 'ℓ' is $\frac{185}{\sqrt{3}}$ kms, then the value of 'x' for which freight charge is minimum :-

(A) $\frac{60}{\sqrt{3}}$ kms (B) $60\sqrt{3}$ kms (C) $120\sqrt{3}$ kms (D) $75\sqrt{3}$ kms

12. If ℓ = 10 kms, then the value of x for which freight charges are maximum :-

(A) 10 kms (B) $10-\frac{5}{\sqrt{3}}$ kms (C) $\frac{10}{\sqrt{3}}$ kms (D) 0 km

Paragraph for Question 13 & 14

If $f : R \to R$ is a function defined by $f(x) = 2\cos^2 2x + \frac{3}{4}\sin 4x + ax$, where $a \in R$.

On the basis of above information, answer the following :

13. Complete set of values of 'a' for which $f(x)$ is strictly increasing for all $x \in R$, is -

(A) $(-\infty, -5]$ (B) $[5, \infty)$ (C) $(-5, \infty)$ (D) $(-\infty, 5]$

14. Complete set of values of 'a' for which $f(x)$ does not have any critical point, is -

(A) $(-\infty, -5]$ (B) $[5, \infty)$ (C) $(-5, 5)$ (D) $(-\infty, -5) \cup (5, \infty)$

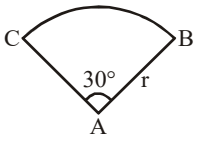
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SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. If m is the slope of a line which is tangent to $y^3 = x^4$ & a normal to $x^2 - 2x + y^2 = 0$, then $\left(\frac{3m}{4}\right)^3$ is equal to ($m \neq 0$)

2. Consider a circular arc ABC as shown in figure, having variable radius r. The circle passing through points A, B, C is having radius R. The value of $\left(2\cos 15° \times \frac{dR}{dr}\right)$ is equal to



3. Water is flowing in a container at a rate proportional to the square of the amount of water present inside the container. If at t = 0, volume of water inside the tank is V_0 & it takes 6 hours for the volume to become $2V_0$, then the time (in hours) taken for the volume to become $4V_0$ is equal to

4. Let $f(x)$ be a function defined by $f(x) = \int_3^x t\left(t^2 - 7t + 12\right)dt$, $3 \le x \le 6$. If the range of $f(x)$ is [a, b], then the value of $\left(3a + \frac{b}{9}\right)$ is

REVISION ASSINGMENT # 14 II & DI-02 MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	A	A	A	A	A	A,C,D	A,B,C,D
	Q.	11	12	13							
	A.	A,C	A,C	B							
SECTION-III	Q.	1	2	3	4	5	6				
	A.	2	5	9	9	2	7				

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